FORM HS-03 · REV. 4 POST-STASIS REFRESHER · MODULE 3 OF 9 POST AT EVERY MUSTER POINT

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

HEAT KILLS.

THIS WORLD IS HOTTER THAN ANYTHING YOUR BODY REMEMBERS. READ EVERY LINE BEFORE YOUR FIRST SHIFT.

JUST WOKE UP?

YOUR HEAT TOLERANCE DID NOT SURVIVE STASIS.

However many seasons you've worked in heat before, **you are a new worker now.** Your body has to rebuild its heat tolerance from zero over 7–14 days. Most heat deaths happen in a worker's **first few days** on a hot site. Report to Med Bay for a weigh-in and a hydration check before your first shift. Wear the orange **RAMP** sticker on your helmet until Day 5.

20%	40%	60%	80%	100%
DAY 1	DAY 2	DAY 3	DAY 4	DAY 5

Share of a full shift's heat exposure you may work. Light duty first. Your crew lead logs your ramp day on the roster. A fever, a stomach bug or a week off-site sends you back to Day 1.

SITE HEAT LEVELS

POSTED AT MUSTER EACH SHIFT AND EVERY HOUR
BY YOUR CREW LEAD · CH 16 BROADCASTS CHANGES

70-79°	**GREEN** Normal ops	Water on your belt. Heavy suits and hard labor can still drop you. **Watch new wakers.**
80-89°	**CAUTION** Heat plan active	Cool water and shade within reach of every crew. 1 cup every 15–20 min. New wakers follow the ramp.
90-102°	**EXTREME CAUTION** Buddy system	At least a **15-min shade break every 2 hours**. Buddies check each other. Crew lead walks the line.
103-124°	**DANGER** Restricted work	Heavy work moves to night shift. Work/rest by the table below. Cooling station on every deck. **Anyone with symptoms is pulled.**

ABOVE THIS LINE: OLD EARTH HEAT CHARTS STOP. RULES FROM HERE ON ARE SITE RULES.

125-139°	**EXTREME** Essential work only	Air is hotter than your skin, so wind heats you and fans don't help. **Cooling vest mandatory.** Pairs only. Exposures of 30 min or less, then a cooled shelter.
140-159°	**HOT WORK** Permit required	Cooled suit or supplied cool air. Entries timed in minutes, logged on the board. Standby rescuer at the hatch. Medical check after every entry.
160-180°	**SITE CLOSED** Emergency only	All crews to shelter. Steel, rebar and sand **burn skin in about one second**. Entry only for rescue, in full reflective cooling gear with a timed return.

Levels up to 124° are heat index in shade. Direct sun adds up to 15°. Your lead also posts the WBGT heat-stress reading (see the work/rest table).

HEAT STROKE

Confused, slurring, stumbling, collapsed or seizing, hot to the touch. **They may still be sweating.** Don't wait to be sure.

**COOL FIRST.
TRANSPORT SECOND.**

Nearly everyone survives if their body temperature is brought under 104° within 30 minutes. Every minute you wait costs them.

1. **Ch 1: "Emergency, emergency, emergency. Heat casualty"**, with sector-grid and deck. The medic is coming to you.
2. **Get them out of the heat.** Strip off suit, vest and helmet.
3. **Dunk them in the ice trough** at the nearest cooling station, up to the neck, holding the head up.
4. No trough? **Soak them**, cover them in ice-water towels, pack ice at the neck, armpits and groin, and fan hard.
5. **No drinks if they're confused.** Stay with them and keep cooling until the medic takes over.

KNOW THE SIGNS

IN YOURSELF AND IN YOUR CREW

HEAT RASH

Clusters of red bumps on the neck, chest or groin.

DO: keep the skin dry, cool off, use powder.

HEAT CRAMPS

Painful spasms in the legs, arms or belly during heavy sweating.

DO: shade, water plus electrolytes. Report it if cramps last over an hour.

FAINTING

Dizziness or blacking out after standing or getting up fast.

DO: sit or lie down somewhere cool and sip water. Tell your lead.

HEAT EXHAUSTION

Heavy sweat, clammy skin, headache, nausea, weakness, short temper, dark urine.

DO: pull off the line, cool down, sip water. Forward Aid Station if not better in an hour. Done for the day.

RHABDO

Muscle pain and weakness, urine dark like tea or cola after hard work.

DO: stop work and go to the Forward Aid Station now. It can wreck your kidneys.

HEAT STROKE

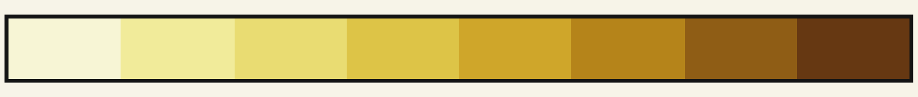
Confusion, slurred speech, collapse, seizures, body at 104° or above.

DO: follow the red box. Cool them on the spot.

WATER DISCIPLINE

8 OZ	48 OZ	2%
EVERY 15–20 MIN	MAX PER HOUR	WEIGHT LOST = PULL OFF

- Drink before you're thirsty. Thirst runs behind your losses.
- Past 48 oz an hour your gut can't keep up, and too much plain water thins your blood salt. Eat the salt tabs or rations with it.
- Weigh in and out at the hatch. Drink 16 oz for every pound lost.
- No alcohol within 24 hrs of a shift. No stim drinks on deck.



WORK / REST BY WBGT

WBGT (wet-bulb globe temperature) is the heat-stress number your crew lead reads off the site meter. Unlike plain air temperature, it counts everything that heats your body: air temperature, humidity, direct sun and wind. That's why a humid 88° WBGT can be more dangerous than a dry 110° air reading.

Each cell shows **minutes of work / minutes of rest per hour**, plus water per hour. These times are for workers who have finished the 5-day post-stasis ramp. **On Days 1–4 of your ramp, work half these times.**

WBGT °F	LIGHT	MODERATE	HEAVY
78–81.9	No limit · ½ qt	No limit · ¾ qt	40/20 · ¾ qt
82–84.9	No limit · ½ qt	50/10 · ¾ qt	30/30 · 1 qt
85–87.9	No limit · ¾ qt	40/20 · ¾ qt	30/30 · 1 qt
88–89.9	No limit · ¾ qt	30/30 · ¾ qt	20/40 · 1 qt
90+	50/10 · 1 qt	20/40 · 1 qt	10/50 · 1 qt

Light: inspections, survey, operating equipment from an AC cab. **Moderate:** framing, tying rebar, carrying panels. **Heavy:** trenching, hauling, shoveling, concrete finishing.

ON THE BUILD

- **Metal burns.** Sun-baked steel, rebar and hull plate pass 158°. Gloves on every time you touch metal.
- **Pours and lifts go at night.** Concrete, welding and heavy rigging get scheduled for the cool hours.
- **Enclosed modules trap heat.** Unfinished habs and hull sections can run far hotter than outside. Check the reading before you go in.
- **Cover up.** Loose, light-colored, full-coverage clothing, a brimmed hard-hat shade, goggles. Bare skin soaks up heat when the air is hotter than you.
- **Wet cloth works in dry air.** A soaked neck wrap cools you through evaporation. It stops working when it's humid.
- **Never sit on bare sand or deck plate.** Use your mat.

CREW LEADS & CONSTRUCTION MANAGERS

- Take the **WBGT reading** at muster and every hour after. Post the level and the work/rest cycle.
- Assign a **buddy pair** to every worker. Check every crew member at least every 30 min.
- Keep **water, shade and a cooling station** no more than a short walk from any work face. Stock an ice trough on every deck.
- Track each new waker's **ramp day**. No one skips the ramp because the schedule slipped.
- Run a **radio check** every 30 min with each crew. A crew that misses one gets checked in person.
- Brief the **heat casualty plan** at every muster: nearest trough, Ch 14 medical, evac route.
- Plan heavy tasks for the night shift. **Cut scope before you cut rest.**

STOP-WORK AUTHORITY

Every worker on this site can stop work when a crew member shows heat symptoms or water, shade or cooling isn't available. No penalty, no questions. A late habitat is better than a dead crew.

WORKER ACKNOWLEDGEMENT

I have read this poster, I know where my nearest cooling station and ice trough are, and I understand that I must follow the post-stasis ramp.

NAME ____________________
COLONIST ID ____________________
CREW / SECTOR ____________________
DATE ____________________

Adapted from Old Earth standards: OSHA & NIOSH heat guidance (2016 criteria), NOAA National Weather Service heat index, US Army TB MED 507, CDC heat illness first aid, Korey Stringer Institute heat stroke protocol.

HS-03 · Dunewright Colony Works · Replace when faded